



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 1C: Introduction to VHDL

Introduction to VHDL

- Initiated in 1981 by the department of defense
- Cost of reproducing hardware as technologies advanced was reaching crisis point
- VHDL: VHSIC Hardware Description Language
 - VHSIC: Very High Speed Integrated Circuits
- VHDL '87 first VHDL Standard
 - Revised in '93, '00, '02, '08, '19
 - '93 is well supported among tool vendors – **primary focus of this course**
 - '00 adds protected types
 - '02 buffer type changes
 - '08 major update with support for PSL, fixed/floating point packages, improved syntax – Not fully supported by all tool vendors though getting much better!
 - '19 adds more language improvements to reduce overall language verbosity – not well supported at this time

VHDL Modeling

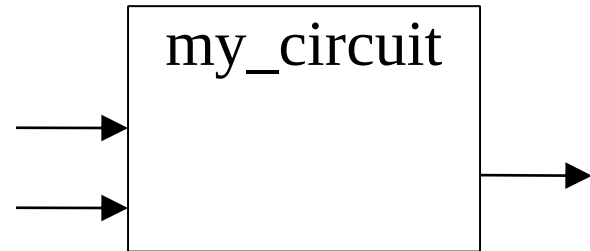


- Digital Circuits are highly parallel
- Simulation for verification
- VHDL provides means to model circuit
- How do we use VHDL to describe a circuit?

VHDL Entity Examples

- Every hardware component has an entity/architecture pair

```
entity my_circuit is port (  
    a : in std_logic;  
    b : in std_logic;  
    c : out std_logic );  
end my_circuit;
```



- Entity describes name of component, ports, types, and directions of ports
 - Ports is an entry into or out of a design component

Port and Signal Values

- IEEE 1164 standard defines 9 values for std_logic:
 - '1' – logic value 1
 - '0' – logic value 0
 - 'U' – uninitialized
 - 'X' – Forcing Unknown
 - 'Z' – High Impedance
 - 'W' – Weak Unknown
 - 'L' – Weak 0
 - 'H' – Weak 1
 - '-' – Don't Care
- Std_logic is sometimes referred to as Metalogical Values

← Simulation Only

Port Direction

- All ports must have an identified direction:

- IN



- OUT



- IN/OUT



- BUFFER



- Buffer is **not** used in this course

Architecture

- Every hardware component has an entity/architecture pair
- Entity describes the component name and ports, while the architecture describes what it does

```
architecture my_architecture of my_circuit is
    -- declarative section (e.g. signals)
begin
    -- activity statements
end my_architecture;
```

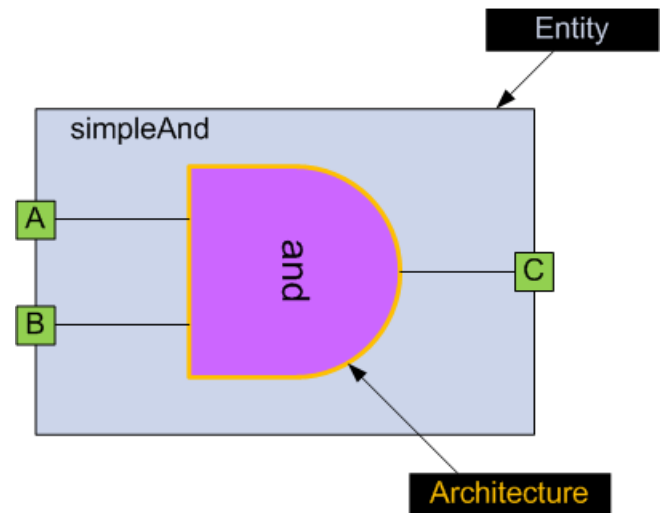
The name of the entity
for which you are
describing the
architecture

- Important: Common practice is to have an entity/architecture pair in a single file

Basic VHDL Design

- Write entity/architecture in a single file

```
library IEEE;  
use IEEE.std_logic_1164.all;  
use IEEE.numeric_std.all;  
use work.all;  
  
entity simpleAnd is Port (  
    A : in std_logic;  
    B : in std_logic;  
    C : out std_logic);  
end simpleAnd;  
  
architecture Behavioral of simpleAnd is  
    signal tbd: std_logic;  
begin  
    C <= A and B;  
end Behavioral;
```



- Save the file as simpleAnd.vhd

Basic VHDL Design Continued

- Write entity/architecture in a single file

```
library IEEE;  
use IEEE.std_logic_1164.all;  
use IEEE.numeric_std.all;  
use work.all;
```

Every design should have this header,
nothing more, nothing less

```
entity simpleAnd is Port (
```

Entity definition

```
    A : in std_logic;  
    B : in std_logic;  
    C : out std_logic);
```

Ports: in, out, inout

```
end simpleAnd;
```

```
architecture Behavioral of simpleAnd is
```

Architecture definition

```
    signal tbd: std_logic;
```

Signal declaration

```
begin
```

```
    C <= A and B;
```

Design function/behavior

```
end Behavioral;
```

- Save that file as simpleAnd.vhd